

NOUR WALID ABDEL-AAL

AI Engineer

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SUMMARY

AI Engineer specializing in Agentic AI, RAG Systems, and Multi-Agent Architectures, with hands-on experience designing and deploying end-to-end intelligent systems across NLP, Speech AI, and Computer Vision. Honors graduate (GPA 3.63/4.0) from October 6 University with a proven ability to deliver production-ready AI pipelines integrating Large Language Models (LLMs), vector databases, and multi-source data — from schema design to interactive dashboards. Native Arabic speaker with strong foundations in Transformer-based ASR (Whisper family) and Arabic NLP, building real-time and offline transcription, summarization, and information-extraction pipelines. Track record of shipping autonomous robotic systems and scalable NLP and Computer Vision solutions with measurable real-world impact.

EXPERIENCE

Agentic AI Developer Intern — NTI — (DEY) Training Program Jan 2026 – Apr 2026

- Architected agentic AI workflows for autonomous task planning, reasoning, and decision-making across multi-step objectives, delivering 3+ validated working systems evaluated and approved by program instructors.
- Engineered Retrieval-Augmented Generation (RAG) systems leveraging LLM pipelines and vector databases, enabling context-aware information retrieval across 5+ heterogeneous data sources, with components for text summarization, NER, and keyword extraction over both English and Arabic content.
- Orchestrated multi-agent architectures enabling collaborative problem-solving and automated knowledge discovery, reducing manual data-processing steps by over 60% compared to baseline workflows.

EDUCATION

B.Sc. in Mechatronics Engineering — October 6 University, Faculty of Engineering Oct 2020 – Jun 2025

GPA: 3.63 / 4.0 — Excellent with Honors — Major: Mechatronics Engineering

SKILLS

LLMs and Agents: LangChain, LangGraph, Groq, ReAct Agents, Multi-Agent Orchestration, RAG Architecture, Multimodal RAG, Graph RAG, Adaptive RAG, Prompt Engineering, LLM Fine-tuning, PEFT, LoRA, Hugging Face Transformers, Vector Databases (Pinecone, Chroma, FAISS, Qdrant)

Speech and Audio AI: OpenAI Whisper, faster-whisper, Hugging Face ASR Pipelines, Wav2Vec2, Arabic Speech Recognition (MSA & Dialectal), Real-time and Offline Transcription Pipelines, Speech-to-Text (STT), Text-to-Speech (TTS), Coqui TTS, gTTS, ElevenLabs API, pyannote (Speaker Diarization), librosa, Audio Preprocessing and Feature Extraction

MLOps and Infrastructure: Docker, Kubernetes, FastAPI, Streamlit, Git, GitHub, Linux, REST API Design, ETL Pipelines, Celery, RabbitMQ, Message Queues, Model Evaluation and Production Monitoring

ML and NLP: PyTorch, TensorFlow, Keras, Scikit-learn, Transfer Learning, Neural Networks, CNNs, RNNs, LSTM, Encoder-Decoder Architectures, Self-Attention, BERT, spaCy, NLTK, Word Embeddings (Word2Vec, GloVe), Text Summarization, Named Entity Recognition (NER), Text Classification, Keyword Extraction, Topic Modeling (LDA, BERTopic), Feature Engineering, Pandas, NumPy

Computer Vision: OpenCV, YOLO, Pillow (PIL), Albumentations, Image Classification, Object Detection, Image Segmentation, Face Detection and Recognition, Real-time Video Analytics

Robotics and Autonomous Systems: ROS2, Gazebo, RViz, SLAM, Nav2, Path Planning, Robot Kinematics and Dynamics, Control Systems, Sensor Fusion

Data Engineering and Visualization: SQL (SQLite, PostgreSQL), MongoDB, Schema Design, Matplotlib, Seaborn, Plotly, Power BI, Exploratory Data Analysis (EDA), Statistical Analysis

Automation and Tools: BeautifulSoup, Scrapy, Selenium, n8n, Workflow Automation

Languages: Python, C, C++, R, MATLAB, Bash

PROJECTS

BI Agent — Agentic Business Intelligence System — [GitHub](#)

- Designed and built a full agentic BI platform connecting SQL and MongoDB databases, with automatic schema crawling and LLM-driven cross-source relationship detection across 2+ database types.
- Engineered a LangGraph-based 5-node agent graph (intent understanding, SQL planner, MongoDB planner, executor, response formatter) with automated self-correction via retry loops, achieving reliable query resolution with zero manual intervention on valid inputs.
- Built an advanced 7-layer charting pipeline covering data normalization, column classification, LLM-driven chart selection, anomaly detection, trend analysis, correlation detection, and Plotly spec generation with one-sentence insight captioning.
- Developed a business-grade Streamlit dashboard featuring KPI cards, interactive Plotly charts, drill-down navigation, and a schema explorer sidebar — stack: Python, LangGraph, LangChain, Groq, SQLAlchemy, PyMongo, FastAPI, Streamlit.

Code-M8 — AI Coding Agent — [GitHub](#)

- Co-developed an AI coding agent capable of autonomous code generation, debugging, and multi-step reasoning across diverse software engineering tasks, supporting end-to-end task completion with minimal manual intervention.
- Designed multi-agent pipelines enabling autonomous planning, execution, and verification of code solutions, cutting manual debugging iterations by over 50% in test scenarios.
- Integrated agentic reasoning loops allowing the system to self-correct on failed executions and iteratively refine solutions until validation passes — stack: Python, LLM APIs, Deep Agents SDK, Prompt Engineering.

Cooperative Dual Robotic Arm System with Vision-Guided Automation

- Designed synchronized control for two PUMA-standard robotic arms using ROS2, microcontrollers, and PLCs for industrial-grade precision.
- Integrated computer vision pipelines with OpenCV and ROS2 nodes for real-time object detection, classification, and pose estimation.
- Implemented ROS2-based multi-robot coordination for cooperative pick-and-place operations and developed automated conveyor-based sorting workflows for smart manufacturing — stack: ROS2, OpenCV, PLCs, Microcontrollers, Python, C++.

CERTIFICATIONS

Machine Learning Specialization — DeepLearning.AI | 2024

Deep Learning Specialization — DeepLearning.AI | 2024 – 2025

Natural Language Processing Specialization — DeepLearning.AI | 2025

ROS2 Nav2 with SLAM and Navigation — Udemy | Jun – Jul 2025

Ultimate RAG Bootcamp — Krish Naik (Udemy) | 2026

LANGUAGES

Arabic: Native (MSA & Egyptian Dialect — directly applicable to Arabic ASR, NLP, and TTS work) | **English:** Professional